

SUMMARY

Final-year Computer Science student who builds and ships full-stack and backend systems end to end, from APIs and databases to test suites and CI/CD. Projects include a statistics-based regression tester for LLM agents and an AI-assisted research app.

COURSE	YEAR	INSTITUTE
Bachelor of Engineering, Computer Science Engineering	Expected Aug 2027	Chitkara University, Himachal Pradesh, India
Related Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, System Design, Computer Networks		

PROJECTS

Full-stack, backend and LLM-tooling projects

github.com/AlfredBateman

AgentProbe Regression Testing for LLM Agents Python, FastAPI, Next.js, PostgreSQL (pgvector), Redis, Docker, GitHub Actions GitHub · Live demo	- Built a regression-detection platform for LLM agents: every case runs many times (10 runs per case in the shipped suites) and is judged on the full trace and tool calls with 10 rule-based judges plus LLM-rubric and consistency judges. - Implemented the statistics behind the verdict (per-case Fisher exact test with Tarone–Holm correction plus a paired sign-flip permutation test, in exact rational arithmetic); in simulation, false regression alarms on unchanged flaky agents fell from 70.6% to 0.9% versus a single-run check. - Shipped a FastAPI + PostgreSQL backend, Next.js dashboard, Typer CLI, and a GitHub Action that comments on the PR and fails the check on a regression; HTTP/MCP/Python adapters with an SSRF guard. - Backed by 1,145 Python tests, 184 web unit tests, and 14 end-to-end specs (97% coverage on core and API) in a CI-to-deploy pipeline on Vercel, Render, and Neon.
StockPilot Stock Screener with NL Query Engine React, Node.js/Express, TypeScript, MongoDB Atlas, Render, GitHub Actions GitHub · Live demo	- Built and deployed a screener over about 100 NIFTY stocks (filter, sort, search, watchlist, 3-way compare) with an Express API and React frontend; a weekday GitHub Actions job refreshes end-of-day Yahoo Finance data into MongoDB Atlas, with a committed snapshot as fallback. - Designed a plain-English query pipeline (typo fixing, offline rule parser, hosted LLM, answer cache) with the deterministic parser as the floor, so search never depends on a network call; every model response is schema-validated, rate limited, and discarded if invalid. - Added a beginner glossary, a Guide drawer, and zero-result suggestions that show which filter to drop. - Wrote a 70-query evaluation harness scoring the parser and the LLM tier, and documented its limits (self-authored expectations) instead of overstating the 69/70 result.
InvestorLens UX Research App with AI Analysis Next.js 16, TypeScript, Prisma, SQLite, Zod, Gemini API GitHub	- Built a full-stack app linking interviews, findings, personas, journey maps, and a recommendations board, with Zod validation in every Server Action, URL-based filters, cross-entity search, and a live dashboard and printable report. - Added an opt-in Gemini feature that extracts up to 15 candidate pain points per transcript, guarded against hallucination: schema validation, verbatim quote checks, invalid finding IDs downgraded, and human approval before a single-transaction save. - Released v1.0.0 with a seeded sample study and a GitHub Actions pipeline (lint, type-check, AI-helper self-check, build).
CanvasLint Canvas 2D Debugger for Chrome DevTools TypeScript, React, Zustand, Vite, Chrome Manifest V3 GitHub · v1.0.0	- Built a Manifest V3 extension that records Canvas 2D calls through prototype wrappers, batches them per animation frame, and relays them over long-lived ports with exponential back-off into a 10,000-slot ring buffer in the service worker. - Built a DevTools panel with a scrubbable timeline that replays the command log, chosen over pixel snapshots for exact replay at lower memory cost. - Released v1.0.0 as an unpacked-install download and documented the known limitations (no render-block parser yet, one capture buffer shared across tabs).

TECHNICAL SKILLS	
Languages	Python, TypeScript, JavaScript, Java, C++, C, SQL
Backend & Data	FastAPI, Node.js, Express.js, REST APIs, PostgreSQL, SQLite, MongoDB, Prisma
Frontend	React, Next.js, Zustand, Chrome Extensions (Manifest V3)
AI / LLM	Gemini API, LiteLLM, MCP, LLM-agent evaluation, output validation (Zod)
DevOps & Testing	GitHub Actions (CI/CD), Docker, Linux, AWS, unit/integration/end-to-end testing
Tools & Core Concepts	Git, GitHub, Data Structures & Algorithms, System Design

CERTIFICATIONS & TRAINING			
AWS	AWS Cloud Practitioner Essentials; AWS Cloud Quest: Cloud Practitioner		AWS Skill Builder
Nutanix	Nutanix Hybrid Cloud Fundamentals (NHCF) – Course Completion		Nutanix
Infosys Springboard	Software Development & Testing Fundamentals; Python Foundation Certification; Programming Fundamentals using Python, Parts 1 & 2		
GitHub – AlfredBateman		AgentProbe – Live demo	StockPilot – Live demo
			Portfolio – Live site